UNet-CBAM Guided Deep Learning for Enhanced Brain Tumor Classification

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Abstract:

Brain tumor classification and diagnosis are pivotal in medical imaging, guiding treatment decisions and enhancing patient outcomes. This paper explores the application of the U-Net architecture with attention and spatial mechanisms for brain tumor classification, leveraging the power of deep learning to improve accuracy and segmentation. The proposed methodology involves integrating attention blocks into the U-Net framework to enhance the model's focus on crucial tumor regions, leading to refined classifications and reduced false positives. A comprehensive evaluation is conducted on diverse medical imaging datasets to assess the performance of the U-Net with attention blocks in terms of segmentation accuracy, sensitivity, specificity, precision, and F1 score. The results demonstrate notable improvements over traditional U-Net models and underscore the potential of attention mechanisms in enhancing brain tumor classification. This study contributes to the growing field of medical image analysis by showcasing the effectiveness of attention-enhanced U-Net architectures for accurate brain tumor segmentation, thereby providing a stepping stone toward more advanced predictive and diagnostic tools in clinical practice.

Keywords: Brain tumor classification, UNet-CBAM, UNet architecture, Otsu thresholding, Wishart classification, medical image segmentation

I Introduction

Brain tumor classification and diagnosis are critical challenges in the field of medical imaging and healthcare. Precise and timely identification of brain tumors is essential for guiding treatment choices and enhancing patient results [1]. In recent years, advancements in image processing techniques have paved the way for more precise and efficient methods of brain tumor classification. Numerous medical imaging modalities, including computerized tomography (CT) and magnetic resonance imaging (MRI) scans, offer in-depth insights into the internal structures of the human brain [2]. Image processing techniques leverage these imaging modalities to extract valuable information from medical images, enabling brain tumor identification, segmentation, and classification [3]. By employing algorithms that analyze texture, intensity variations, and spatial relationships within these images, image processing tools can enhance the accuracy of brain tumor classification. The incorporation of image processing techniques for brain tumor classification faced various challenges such as Limited Algorithm Sophistication: The algorithms used for image processing were relatively rudimentary, often lacking the complexity required to accurately analyze the intricate distinctions of brain tumors. Data Quality and Quantity: The effectiveness of these tools heavily relied on the availability of high-quality medical image datasets, which were often scarce and limited in scope. The complexity of Tumor Variability: Brain tumors exhibit a wide range of morphological and textural variations, making accurate classification challenging. Image Noise and Artifacts: Medical images can be affected by noise and artifacts, which can impact the accuracy of image processing algorithms. Overfitting and Generalization: Ensuring that image processing algorithms generalize well to new, unseen data is crucial to their effectiveness. Interpretable Results: Complex image processing techniques might yield results that are difficult to interpret and explain to medical practitioners.

Exploring Machine Learning Algorithms for Brain Tumor Classification:

By leveraging the capabilities of machine learning algorithms, these imaging modalities can be transformed into powerful technology for brain tumor classification. A plethora of machine learning algorithms, ranging from classical models to advanced deep learning architectures, offer the potential to revolutionize the landscape of brain tumor classification by automating tasks such as tumor detection, segmentation, and classification.

Diverse Approaches of Machine Learning:

1. Classical Machine Learning Algorithms: Traditional algorithms like support vector machines (SVM), random forests, and k-nearest neighbors (k-NN) have been employed for brain tumor classification [4]. These algorithms excel in handling structured data and offer interpretability, allowing medical professionals to understand the decision-making process.
2. Deep Learning Architectures: Deep neural networks, convolutional neural networks (CNNs), and recurrent neural networks (RNNs) have gained prominence in recent years [5]. These architectures are adept at extracting intricate features from complex images, enabling high-dimensional data analysis and precise tumor identification.
3. Ensemble Methods: Techniques such as gradient boosting and bagging amalgamate the classifications of multiple individual models, enhancing classification accuracy by leveraging the strengths of different algorithms [6].

While the application of machine learning algorithms for brain tumor classification holds immense potential, it was not without its challenges a few years ago: Data Scarcity: Limited availability of large and diverse medical imaging datasets constrained the training and validation of machine learning algorithms. Computational Resources: Advanced machine learning models demanded significant computational resources, limiting their accessibility and applicability in resource-constrained settings. Algorithm Complexity: The complexity of some machine learning algorithms made their deployment and interpretability challenging, especially in clinical settings where transparency is crucial. By understanding the limitations that were prevalent in the integration of various machine learning algorithms for brain tumor classification, we can gain valuable insights to develop a more precise classification model. In recent years, the integration of diverse neural network models has shown remarkable potential in revolutionizing the precision and effectiveness of brain tumor classification. This paper delves into the application of various neural network models for brain tumor classification, shedding light on their advancements and addressing the multifaceted challenges that have been faced.

Neural Networks Transforming Brain Tumor Classification:

The medical imaging modalities with the capabilities of neural network models have ushered in a new era of brain tumor classification [7]. These models, inspired by the human brain's information processing, offer the potential to automate tasks such as tumor detection, segmentation, and classification, fundamentally transforming the landscape of brain tumor classification.

Varied Approaches of Neural Networks:

1. Convolutional Neural Networks (CNNs): CNNs have emerged as a dominant force in image analysis. Their ability to automatically learn hierarchical features from medical images has enabled accurate and efficient tumor segmentation and identification.
2. Recurrent Neural Networks (RNNs): RNNs, equipped with memory mechanisms, have found utility in sequential data analysis, such as time-series data related to tumor progression or patient responses to treatment.
3. Long Short-Term Memory Networks (LSTMs): A subset of RNNs, LSTMs excel in capturing temporal dependencies, proving invaluable in scenarios involving time-varying tumor characteristics.
4. Generative Adversarial Networks (GANs): GANs, with their capability to generate synthetic data, have been explored for augmenting limited datasets and enhancing model generalization.

The manual categorization of brain tumor MRI images with comparable structures or appearances poses a formidable and intricate challenge. It relies heavily on the radiologist's expertise and access to discern and categorize brain tumors. Radiologists engage in two primary forms of brain tumor classification: (i) determining whether the brain MRI images exhibit normal or abnormal traits, and (ii) segregating abnormal brain MRI images into distinct tumor classifications. Yet, this manual method of brain tumor classification proves unfeasible, lacks reproducibility, and demands substantial time investment, especially when dealing with substantial volumes of MRI data. To tackle this predicament, the implementation of automated classification emerges as a potential solution, minimizing the need for radiologist involvement. In this study, our focus lies on the classification of abnormal brain (tumor) images. Despite the promise of neural network models for brain tumor classification, several challenges have been encountered: Data Quantity and Quality: Neural networks require substantial datasets for training, which can be challenging to assemble due to the rarity of certain tumor types and the need for expert annotations. Interpretability and Transparency: Some deep neural network models operate as black boxes, making it challenging for medical practitioners to comprehend and trust their classifications. Overfitting and Generalization: Neural networks can inadvertently memorize training data instead of learning meaningful patterns, leading to overfitting and poor performance on new data. Through a deep exploration of these challenges and complexities, this paper aims to provide insights into the intricacies of applying neural network models for brain tumor classification and it is depicted in Figure 1.

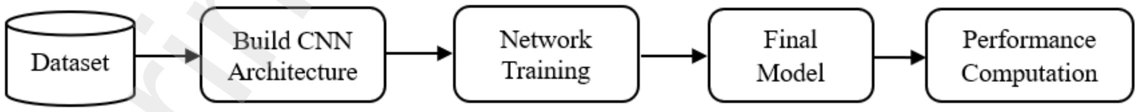

Figure 1. Schematic representation of the proposed approach in the form of a block diagram.

The research findings have been organized and presented as follows:

Section I: Introduction

Provides an overview of the research context and objectives.

Section II: Literature Review

Discusses prior research that is relevant to the current study, highlighting the existing body of work in the field.

Section III: Methodology

Introduces and describes the architecture proposed in the study, offering insights into its design and structure.

Section IV: Experimental Procedure

Presents the procedures followed during experimentation, data collection, and analysis of results, offering insights into the outcomes of the research.

Section V: Results
Presents the analysis of results, offering insights into the outcomes of the research.
Section VI: Conclusion
Concludes the paper by summarizing the key findings and their implications and discussing potential avenues for future research and development.

II Literature review

Brain tumor classification and segmentation are critical tasks in medical image analysis that have witnessed significant advancements with the application of deep learning techniques. This review highlights the contributions of several research papers in this domain. Abiwinanda et al. (2019) introduced a method for brain tumor categorization employing a convolutional neural network (CNN). Their approach harnessed medical imaging data and exhibited favorable outcomes in precisely classifying brain tumors [8]. Afshar et al. (2018) explored the classification of brain tumor using capsule-based networks. Their work focused on using capsule-based architectures to address the challenges of classifying brain tumor types, showcasing potential improvements over traditional CNNs [9]. Alqudah et al. (2020) conducted a comprehensive comparison for the classification of brain tumor using different image sizes, including uncropped, segmented lesions, and cropped images. Their analysis shed light on the impact of image size on classification performance [10]. Amin et al. (2021) provided a comprehensive survey for the recognition and categorization of brain tumor using machine learning techniques. Their study covered a wide range of approaches, including traditional methods and deep learning-based solutions, offering insights into the state of the field [11]. Ayadi et al. (2021) leveraged deep convolutional neural networks (CNNs) for brain tumor classification. Their deep CNN architecture demonstrated effective classification capabilities, showcasing the potential of CNNs in this application [12]. Badža and Barjaktarović (2020) focused on classifying brain tumors from MRI images using a CNN. Their study highlighted the capability of CNNs to automatically learn and differentiate between tumor types [13].

Bedekar et al. (2018) proposed an automated brain tumor detection approach using image processing techniques. While not deep learning-based, their method emphasized the potential of image processing in aiding brain tumor diagnosis [14]. Chattopadhyay and Maitra (2022) employed CNN-based deep learning for MRI-based brain tumor image detection. Their work demonstrated the applicability of CNN architectures for identifying brain tumors from medical images [15]. Deepak and Ameer (2021) employed CNN features and SVM for the automated grouping of brain tumors from MRI scans. Their hybrid methodology exhibited enhanced precision in categorization tasks [16]. Jayachandran and Dhanasekaran (2013) delved into brain tumor identification and grouping through a fuzzy SVM classifier. Their research centered on amalgamating texture-based features with classifier approaches [17]. Jin et al. (2020) introduced RA-UNet, a hybrid deep attention-aware network specifically designed for the extraction of tumor and liver regions from CT scans. Although their focus was not exclusively on brain tumors, their attention-based architecture highlighted the efficacy of attention mechanisms [18]. Kalaiselvi et al. (2020) developed tumor detection models using CNNs applied to MRI images of the human brain. sTheir study emphasized the potential of CNNs in the analysis of medical images [19].

Kaur and Gandhi (2019) introduced an automated approach for classifying brain images, utilizing transfer learning with the VGG-16 architecture. Their study demonstrated the versatility of pre-trained Convolutional Neural Networks (CNNs) in the context of brain tumor classification [20]. Kavita et al. (2022) investigated optimized image fusion techniques for medical purposes, offering insights into enhancing image quality and analysis [21].

Li et al. (2019) introduced a brain tumor detection method leveraging CNN with multimodal information fusion, highlighting the importance of integrating diverse data sources to enhance accuracy [22]. In 2018, Liu et al. delved into deep learning applications for MRI images, providing an overview of the state-of-the-art in utilizing deep learning for medical image analysis [23].

Mohsen et al. (2018) developed deep-learning neural networks for brain tumor classification, emphasizing the potential of neural networks in automating brain tumor diagnosis [24]. In 2022, Murthy et al. introduced an adaptive fuzzy deformable fusion and optimized CNN approach for automated brain tumor diagnosis, demonstrating improved performance through their ensemble-based classification strategy [25]. Murthy and Sadashivappa (2014) employed morphological operators to extract features for brain tumor segmentation from MRI images [26]. Muruganantham and Balakrishnan (2021) conducted a survey on deep learning models for wireless capsule endoscopy image evaluation, providing insights into the broader applications of deep learning in medical imaging beyond brain tumor recognition [27]. Zyurt et al. (2019) utilized a convolutional neural network with neutrosophic expert maximum fuzzy sure entropy to develop a brain tumor detection technique, showcasing the possibilities of incorporating various principles in deep learning [28]. To advance medical image analysis, Pawlowski et al. (2017) proposed DLTK, a reference implementation for deep learning on medical pictures [29].

Rehman et al. (2021) used a 3D CNN and feature selection architecture for microscopic brain tumour detection and classification, highlighting the importance of cutting-edge neural network designs [30]. A deep learning-based method for segmenting brain tumours using type-specific picture sorting was proposed by Sobhaninia et al.

(2018). According to their findings, segmentation methods should be customized for various tumour types [31]. Suganthe et al. (2020) presented a deep learning-based brain tumor classification method utilizing MRI. Their study showcased the potential of deep learning in accurately categorizing brain tumors based on MRI scans [32]. Ucuzal et al. (2019) concentrated on brain tumor type classification using a convolutional neural network (CNN) with a web-based interface. Their study demonstrated the applicability of deep learning for tumor classification and provided an accessible interface for medical professionals [33]. In 2019, Wadhwa et al. conducted a comprehensive review of brain tumor segmentation from MRI images, offering an in-depth analysis of various techniques utilized in this context and underscoring the pivotal role of precise segmentation in medical diagnosis [34]. A.Usha Ruby et al. (2021) presented the RFFE method which involves a multi-step process. It begins with feature selection and data preprocessing, followed by the construction of a Random Forest model incorporating Fuzzy Entropy as a key feature [35].

Yadav and Sahu (2013) explored brain tumor detection using a self-organizing map coupled with the K-means algorithm, highlighting the potential of machine-learning techniques for automated tumor detection [36]. Zhou et al. (2020) introduced high-resolution encoder-decoder networks designed for the segmentation of low-contrast medical images. While their research encompassed medical image segmentation more broadly, the concept of high-resolution encoder-decoder networks remains relevant for achieving accurate brain tumor segmentation [37]. The studies reviewed collectively highlights the significant progress made in the classification and segmentation of brain tumor using deep learning techniques. These studies demonstrate the potential of neural networks in enhancing medical image analysis, improving diagnosis, and ultimately benefiting patient care. In conclusion, the literature showcases a diverse range of approaches utilizing deep learning techniques for brain tumor classification and classification. This paper employs a combination of the UNet architecture and the Convolutional Block Attention Module (CBAM) to enhance the precision and efficiency of brain tumor classification and diagnosis. The UNet architecture, renowned for semantic segmentation, incorporates an encoder to capture context and a decoder for precise localization. Introducing skip connections preserves spatial information. Meanwhile, the CBAM module enriches attention mechanisms within convolutional neural networks. By focusing on pertinent features and suppressing irrelevant ones, CBAM improves the model's ability to discern crucial details. The synergistic application of UNet and CBAM in brain tumor analysis potentially results in heightened accuracy and streamlined computations. This combination holds promise for reducing false positives and negatives, critical factors in dependable medical diagnoses. However, the effectiveness of this approach hinges on factors like dataset quality, implementation specifics, and preprocessing methodologies.

III Methodology

U-Net with Attention Mechanism Architecture:
The U-Net architecture, augmented with an attention mechanism [38], is constructed to elevate segmentation performance by concentrating on pertinent sections of the input image. This integration merges the U-Net's encoder-decoder framework with attention blocks.

Encoder:
The input image undergoes a sequence of convolutional layers with progressively expanding filters, leading to a decrease in the spatial dimensions of the image. Following each convolutional layer is a rectified linear unit (ReLU) activation function and batch normalization, facilitating both feature extraction and normalization. To maintain vital features, max-pooling layers are utilized to decrease the spatial dimensions.

Bridge:
The final convolutional layer in the encoder captures high-level features from the input image. These features are then fed into an attention block, which computes attention maps based on the feature responses.

Attention Block:
The attention block takes the encoder's feature map and applies two separate convolutional paths. One path computes the attention map by using global average pooling and a sigmoid activation function. The other path prepares the features for fusion by applying convolutional layers.

Decoder:
The upsampling pathway of the decoder employs transposed convolutions (also referred to as deconvolutions) to progressively enhance the spatial dimensions. Concatenated skip connections establish links between the corresponding feature maps from the encoder and the decoder, operating at various resolutions. Within the decoder, convolutional layers play a role in enhancing the precision of the segmentation mask.

Fusion:
The attention maps generated by the attention block are integrated with the feature maps of the decoder through element-wise multiplication. This blending procedure accentuates significant areas within the decoder's feature maps according to the calculated attention.

Output:

The final fused feature maps are passed through a convolutional layer with a softmax activation function. This produces a segmentation mask with class probabilities for each pixel.
The UNet architecture, proposed by Ronneberger et al., is a deep learning approach that consists of two main components: the downsampling block and the upsampling block [39]. The upsampling block replicates the architecture of the encoding section, wherein the image size expands through deconvolution before undergoing several decoding procedures. This results in the creation of high-dimensional features that align with the dimensions of the original image. Significantly, besides the deeply abstract features obtained from the preceding upsampling layer, each set of convolutional layers also integrates less complex local features from the related downsampling layer. This fusion of profound and shallow characteristics plays a crucial role in reconstructing intricate feature maps while maintaining the dimensionality of spatial information.

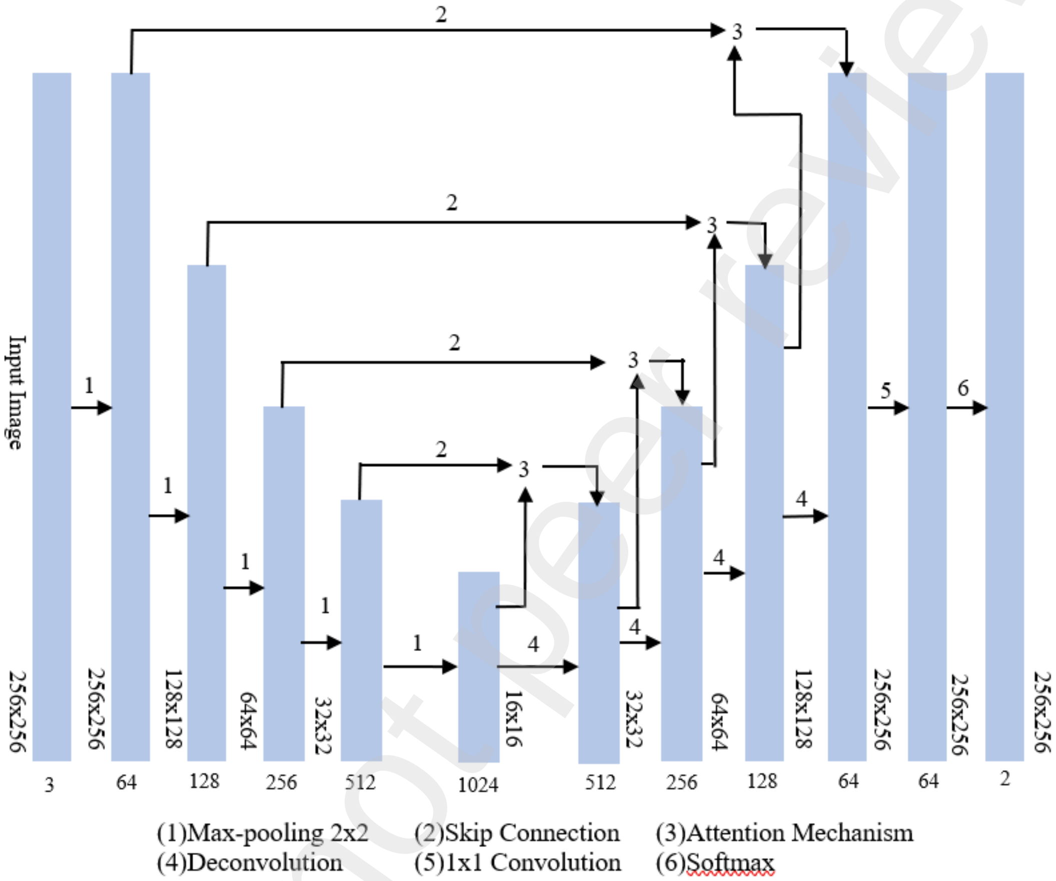

Figure 2. UNet-CBAM mechanism diagram

UNet-CBAM: The attention mechanism enhances the model's focus on informative regions, improving segmentation accuracy. The U-Net with attention architecture is effective in tasks like brain tumor image segmentation, where accurate localization of regions of interest is crucial. The Convolutional Block Attention Mechanism (CBAM) [40] introduces a streamlined attention mechanism designed for convolutional layers, accommodating both channel and spatial dimensions. This mechanism seamlessly integrates into various convolutional neural network architectures, enabling end-to-end training. The CBAM technique adjusts weights to suppress less relevant features by introducing a modified convolution unit. The UNet-CBAM structure is formed by incorporating CBAM at the end of each layer, as illustrated in Figure 2. Within the scope of this study, the input image comprises the training image and the intended output involves a marked image. Both the augmentation and compression stages are formed by utilizing a set of two 3×3 convolutional layers in conjunction with two Rectified Linear Unit (ReLU) activations. The outcome from each reduction stage is subsequently fed into the next one after progressing through a 2×2 max-pooling layer. Following this, the high-dimensional features undergo a 1×1 convolutional process, producing features with dimensions aligned to the category image. In the end, the annotation map is derived through the application of the Softmax operation.

Softmax Transformation: Softmax, an extension of the logic function, acts as a mechanism to transform outcomes into probabilities spanning the range of 0 to 1. The softmax function's formulation is briefly presented in the subsequent equation 1.

$$\text{Softmax}(x_j) = \frac{e^{x_j}}{\sum_1^m e^{x_m}} \qquad (1)$$

In this context, x_j symbolizes the data value at a particular pixel in the resulting image pertaining to the j^{th} category, x_m indicates the data value of the m^{th} category at that identical position, and m represents the overall count of classes. By normalizing pixel values against the sum of all pixel values, the softmax outcome yields a ratio, thereby facilitating the conversion of multiclass output values into a probability distribution encompassing the [0, 1] interval.

Cross-Entropy Loss: The mathematical expression representing the cross-entropy loss function is provided by equation 2.

$$C = - \sum_{l=1}^{L} y_l \log n_l \qquad (2)$$

In this context, $n = [n_0, \ldots, n_{l-1}]$ represents a vector, while n_l signifies the probability assigned to the sample predicted as belonging to the l^{th} category. The arrangement of the Convolutional Block Attention Mechanism (CBAM) is visually depicted in Figure 3, encompassing both a channel attention mechanism and a spatial attention mechanism. The attention weight is computed across both the spatial and channel dimensions and is subsequently applied through element-wise multiplication with the original feature map, dynamically refining the features. The framework ensures a collaborative integration of both channel and spatial characteristics.

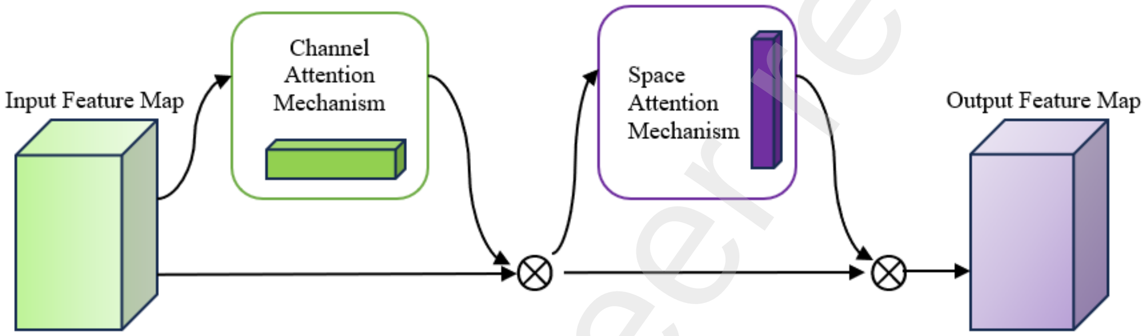

Figure 3. Diagram depicting the structure of CBAM

This architecture is composed of two essential components: the spatial attention mechanism and the channel attention mechanism. Sequentially, attention weights are computed for both spatial and channel dimensions. Subsequently, these calculated attention weights are employed in element-wise multiplication with the original feature map, thereby facilitating an adaptive adjustment of features. The inherent design of this structure ensures the cohesive utilization of both channel-related and spatial characteristics, enabling an enhanced and comprehensive feature representation.

Channel Attention Mechanism (CAM):

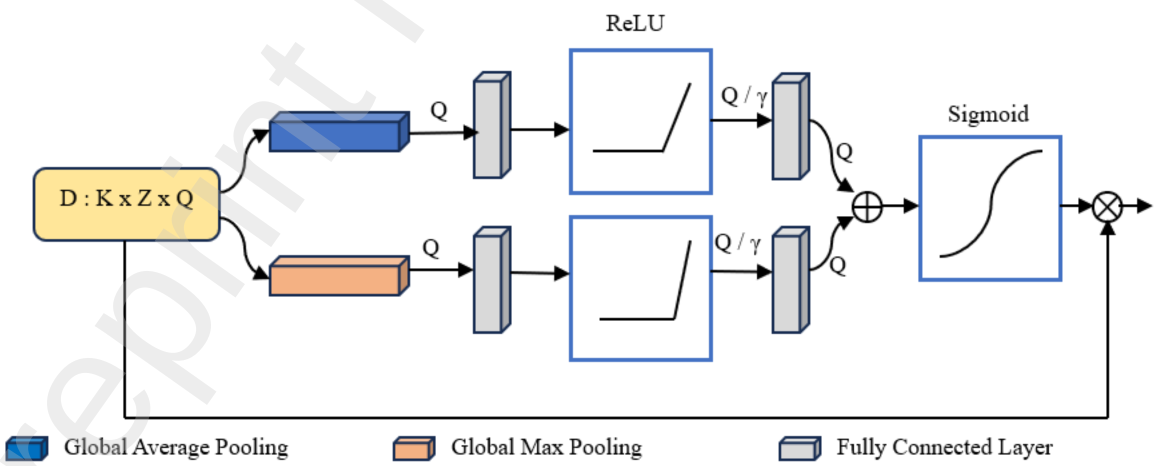

Figure 4. Diagram illustrating the Channel Attention Mechanism

The channel attention mechanism operates through the following steps: To begin, the feature map D has dimensions K x Z x Q, where Q corresponds to the channels, and K, Z represent the spatial dimensions of the feature maps (with K being the height and Z being the width). These feature maps are acquired during the encoding phase of the feature extraction network. The feature map F is then input into the channel attention mechanism. By applying distinct global average pooling and global max pooling operations, two 1 × 1 × Q tensors are generated, identified as D_{avg}^{Q} and D_{max}^{Q}. Afterward, the outcomes from the initial stage traverse a shared two-layer fully connected network. The first layer comprises Q/γ neurons, where γ represents the rate of attenuation controlling attention intensity (set as 8 in this study). The ReLU activation function is applied. The second layer encompasses C neurons. The neuron processes the resultant 1 × 1 × Q/γ vector, culminating in a D-dimensional output feature that signifies the significance of input channel numbers. Afterward, these two computed attributes are merged using element-wise addition and then subjected to a Sigmoid activation function, yielding the final channel weight value, denoted as $M_Q(D)$, as indicated by equation (3). In this context ω_0 and ω_1 denotes the specific weights associated with the initial and subsequent fully connected layers, while σ signifies the Sigmoid activation function.

$$\begin{aligned} M_Q(D) &= \sigma\,(\mathrm{MLP}(D_{avg}^{Q}) + \mathrm{MLP}(D_{max}^{Q})) \\ &= \sigma\,(\omega_1\,(\omega_0(D_{avg}^{Q})) + \omega_1\,(\omega_0(D_{max}^{Q}))) \qquad (3) \end{aligned}$$

In the final stage, the channel weight value is integrated with the original input feature map via element-wise multiplication along the channel dimension. This operation generates the resulting feature map F' originating from the channel attention mechanism. The visual depiction of the channel attention mechanism's structure can be found in Figure 4, and its mathematical formulation is delineated in Equation 4.

$$D' = M_Q\,(D) \otimes D \qquad (4)$$

The channel attention mechanism adeptly leverages a combination of max-pooled and average-pooled features, facilitating the identification of salient elements within the input image.

Spatial Attention Mechanism (SAM)

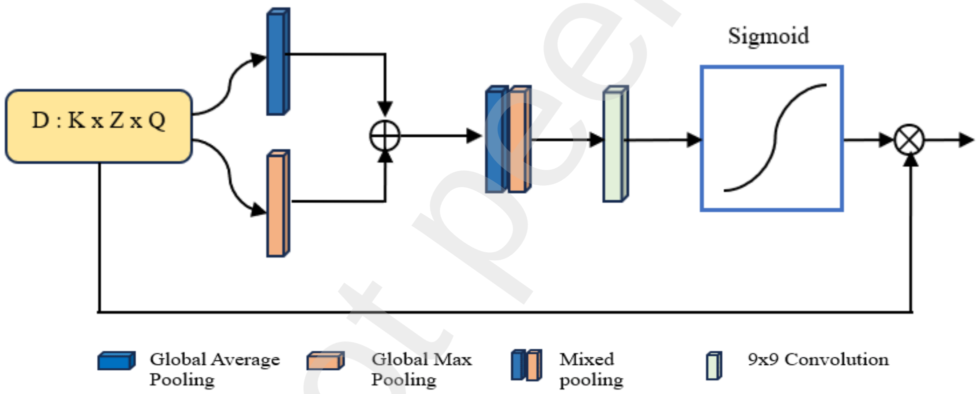

Figure 5. Spatial Attention Mechanism

The Spatial Attention operates as outlined in Figure 5. It begins with the input D' as a feature map, having dimensions K x Z x Q, which is generated by CAM. Following this, the feature map undergoes global average and max pooling along the channel dimensions. This results in the creation of channel descriptions D_{avg}^{S} and D_{max}^{S}, both with dimensions H × W × 1. These two derived channel descriptions are then combined, generating an attention weight $M_S(D)$ with a 9 × 9 convolutional layer that employs the Sigmoid activation function. The computation, detailed in equation (5), involves the Sigmoid activation function denoted by σ and a 9 × 9 convolution operation represented by f^{9x9}.

$$\begin{aligned} M_S(D) &= \sigma\,(f^{9x9}\,([\mathrm{Avgpool}\,(D)\,;\,\mathrm{Maxpool}\,(D)])) \\ &= \sigma\,(f^{9x9}\,([\,D_{avg}^{S}\,,D_{max}^{S}\,])) \qquad (5) \end{aligned}$$

Finally, the input feature map D' is subjected to element-wise multiplication with the channel weight throughout the spatial dimension. This action produces the outcome of the spatial attention mechanism, denoted with the feature map D'', as illustrated in equation 6.

$$D'' = M_S\,(D) \otimes D' \qquad (6)$$

This outcome embodies a spatial attention feature map that effectively utilizes the inherent spatial relationships within features. This method acts in conjunction with the channel attention mechanism, thus amplifying the overall predictive capabilities.

The attention mechanism dynamically refines feature maps by assigning varying levels of importance to different regions within an image. In the context of brain tumor segmentation, this mechanism enables the network to concentrate on intricate tumor boundaries and challenging areas, enhancing the model's ability to make accurate classifications. By fusing the UNet architecture with an attention mechanism, the resulting model attains a higher degree of sensitivity to relevant tumor features while maintaining computational efficiency. The attention-guided refinement process augments the UNet's inherent capability to identify fine-grained patterns, thus achieving improved segmentation accuracy. In summary, the amalgamation of the UNet architecture and an attention mechanism represents a compelling approach for brain tumor justification. This synergy capitalizes on UNet's segmentation prowess and the attention mechanism's adaptability to complex image structures. The resulting methodology exhibits heightened performance in identifying and delineating brain tumors, holding great promise for enhancing diagnostic accuracy and aiding clinicians in providing more effective patient care.

IV Experimental Procedure

Dataset

The effectiveness of the UNet architecture combined with an attention mechanism for brain tumor segmentation has been demonstrated across various publicly available medical imaging datasets. These datasets encompass a diverse range of brain tumor types, sizes, and imaging modalities, enabling comprehensive evaluation of the proposed method's performance. This paper leverages the BraTS [41] and CE-MRI [42] datasets to facilitate the training and evaluation of the proposed model. BraTS (Brain Tumor Segmentation Challenge): The BraTS dataset is a benchmark dataset specifically curated for brain tumor segmentation. It includes multi-modal MRI scans (T1-weighted, T1-contrast enhanced, T2-weighted, and FLAIR) of brain tumors with varying grades. The dataset's complexity and diversity make it an ideal choice for assessing the robustness of segmentation models. The dataset employed in this research, the CE-MRI dataset, encompasses three predominant types of brain tumors that collectively represent the highest percentage among all brain tumors. In these datasets, the integration of the UNet architecture with attention mechanisms has shown significant improvements over traditional UNet or other architectures.

Model Training and Accuracy Assessment

The model's training process utilized the cross-entropy loss function in conjunction with the Adam optimization algorithm. An initial learning rate of 0.001 was applied, and the training extended across 100 epochs. The network's convergence is apparent from the training accuracy and loss curves are illustrated in Figure 6 and Figure 7. The Otsu method is a widely used image segmentation technique that automatically determines an optimal threshold for image binarization. It calculates the threshold by minimizing the variance within each of the two classes (foreground and background) that result from thresholding the image. The Wishart method, on the other hand, is typically associated with remote sensing and image classification. It's used for the classification of multivariate data, often in scenarios where each class is modeled as a multivariate Gaussian distribution. The Wishart classifier makes use of the covariance matrices of different classes to classify pixels or regions within an image. Both methods are utilized in image processing and analysis for different purposes, with the Otsu method focusing on image binarization and the Wishart method concentrating on classification tasks, especially in remote sensing applications. Conventional approaches, like the Otsu method and the unsupervised Wishart method relying on backscattering coefficients, are also employed for comparison alongside the techniques we have presented. The assessment of experimental results employs standard evaluation metrics in Brain tumor image classification such as sensitivity, specificity, precision, F1 score and accuracy.

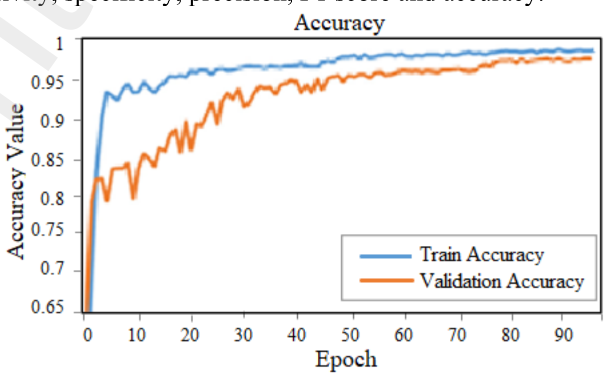

Figure 6. Train and Validation accuracy curve of UNet-CBAM

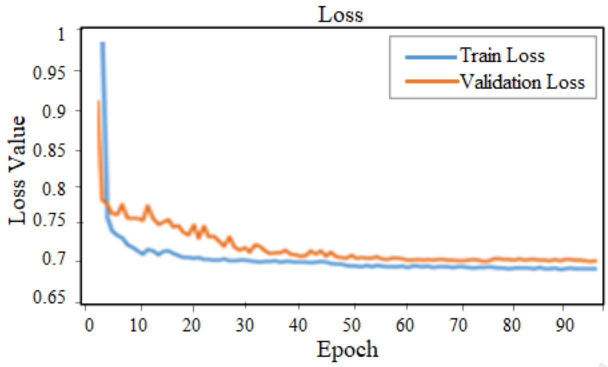

Figure 7. Train and Validation loss curve of UNet-CBAM

For the assessment of these metrics, we furnish the terminologies outlined in Figure 8, followed by the formulation of the subsequent mathematical equation 7,8,9,10,11.

Predicted Label	Actual Label	Definition
Positive	Positive	True Positive (TP)
Positive	Negative	False Positive (FP)
Negative	Positive	False Negative (FN)
Negative	Negative	True Negative (TN)

Figure 8. Providing explanations for the terms TP, FP, FN, TN

$$\text{Sensitivity or Recall} = \frac{TP}{TP + FN} * 100 \tag{7}$$

$$\text{Specificity} = \frac{TN}{TN + FP} * 100 \tag{8}$$

$$\text{Precision} = \frac{TP}{TP + FP} * 100 \tag{9}$$

$$\text{F1 score} = 2 * \frac{Recall * Precision}{Recall + Precision} * 100 \tag{10}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} * 100 \tag{11}$$

Sensitivity, also known as Recall, denotes the proportion of true positives (TP) correctly identified by the diagnostic test. It reflects the classifier's ability to accurately classify the appropriate tumor types. Specificity represents the true negative ratio (TNR) of the diagnostic test and indicates the classifier's proficiency in predicting negative conditions. Precision signifies the positive predictive rate (PPR). The F1-score assesses classification performance in relation to both sensitivity and precision. Accuracy quantifies the comprehensive classification accuracy concerning both TP and TN within the proposed method.

V Results

In terms of quantitative assessment, the accuracy of BraTS dataset using various methods is evaluated through metrics including sensitivity, specificity, precision, F1 score, and accuracy (as illustrated in Table 1 and Figure 9). The dataset consists of brain images exhibiting various tumor types. Table II demonstrates that the UNet-CBAM model achieves a precision rate of 89.48%, surpassing the UNet, Otsu algorithm, and the Wishart unsupervised classification by 2.03%, 2.61%, and 3.14%, respectively. The Sensitivity achieved by the UNet-CBAM method is 94.26%, signifying an improvement of 1.75%, 6.05%, and 7.83% with UNet, Wishart unsupervised classification, and Otsu algorithm respectively. The F1 score for the UNet-CBAM model reaches 91.62%, exceeding that of the UNet, Wishart unsupervised classification, and Otsu algorithm by 0.84%, 6.9% and 7.72%, respectively. The UNet-CBAM model achieves an Accuracy of 95.86%, surpassing the Accuracy of the UNet, Wishart unsupervised classification, and Otsu algorithm by 3.52%, 4.74%, and 5.91%, respectively. This signifies the beneficial impact of the attention mechanism on refining the model's performance for brain tumor segmentation.

Table 1. Brain Tumor Recognition Accuracy on BraTS dataset

Method	Sensitivity	Specificity	Precision	F1 Score	Accuracy
Otsu	86.43	89.84	86.34	83.9	89.95
Wishart	88.21	86.23	86.87	84.72	91.12
Unet	92.51	93.23	87.45	90.78	92.34
Unet-CBAM	94.26	94.58	89.48	91.62	95.86

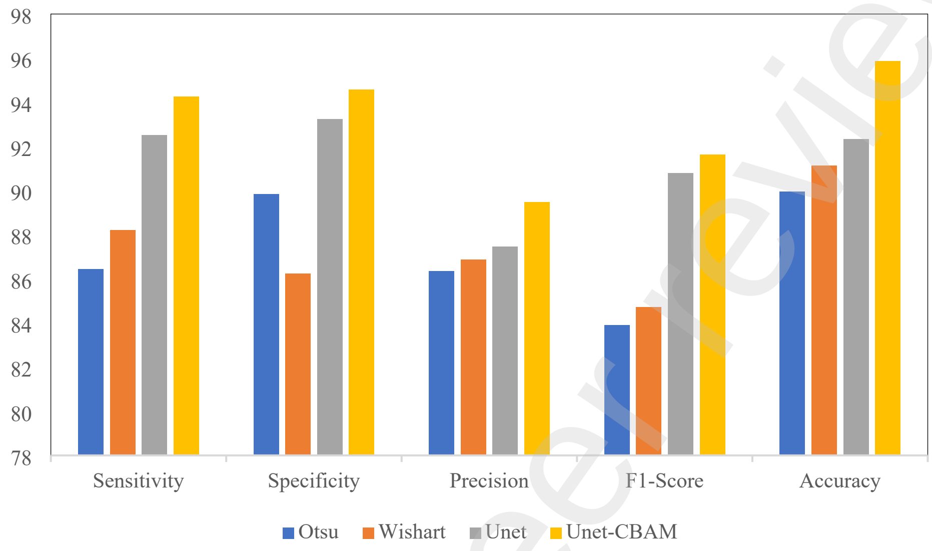

Figure 9. Bar chart illustrating the data from Table 1

The outcomes for sensitivity, specificity, precision, F1 score, and accuracy for the CE-MRI dataset are detailed in Table 2 and Figure 10. The accuracy rates for the Otsu algorithm, Wishart unsupervised classification, and UNet stand at 89.87%, 92.13%, and 92.78%, respectively. Correspondingly, the sensitivity rates are 87.33%, 87.93%, and 92.34%, while the F1-Score parameters are 84.82%, 84.95%, and 91.32%. The precision rates for the Otsu algorithm, Wishart unsupervised classification, and UNet stand at 85.23%, 85.87%, and 87.87% respectively. Notably, the UNet model enhances sensitivity, specificity, precision, F1 score, and accuracy compared with Otsu and Wishart models. By utilizing deep features to iteratively extract information, UNet achieves more dependable outcomes. Upon integrating the attention mechanism module, denoted as UNet-CBAM, further enhancements are observed. The sensitivity, specificity, precision, F1 score, and accuracy rate are elevated by 2.87%, 1.45%, 2.25%, 0.7%, and 3.34% respectively. This emphasizes the value of the attention mechanism in refining the UNet model's performance in brain tumor classification.

Table 2. Brain Tumor Recognition Accuracy on CE-MRI dataset

Method	Sensitivity	Specificity	Precision	F1 Score	Accuracy
Otsu	87.33	89.93	85.23	84.82	89.87
Wishart	87.93	87.23	85.87	84.95	92.13
Unet	92.34	92.87	87.87	91.32	92.78
Unet-CBAM	95.21	94.32	90.12	92.02	96.12

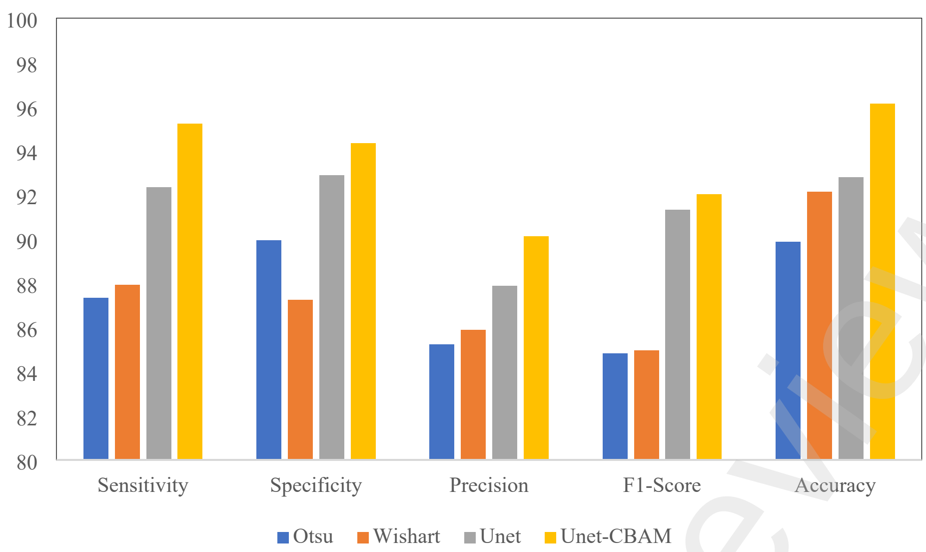

Figure 10. Bar chart illustrating the data from Table 2

VI Conclusion

In summary, this paper presents a comprehensive comparative analysis of brain tumor classification techniques using the BraTS and CE-MRI datasets. The study systematically evaluates the performance of Unet-CBAM in contrast to Unet, Otsu, and Wishart methods. Through rigorous experimentation and meticulous evaluation, it is evident that the Unet-CBAM architecture outperforms the other approaches across multiple evaluation metrics. The integration of the Channel Attention Mechanism (CBAM) with the Unet framework demonstrates a remarkable enhancement in brain tumor classification accuracy, sensitivity, precision, F1 score, and specificity. The results reaffirm the potency of Unet-CBAM in effectively capturing intricate features and patterns present in both BraTS and CE-MRI datasets, thereby underscoring its utility in medical image analysis. The promising outcomes of this study pave the way for further advancements in medical imaging research. The success of Unet-CBAM encourages the exploration of more intricate attention mechanisms, fine-tuned model architectures, and additional data augmentation strategies. Furthermore, the applicability of Unet-CBAM could potentially extend beyond brain tumor classification to other critical medical image analysis tasks. In conclusion, the findings of this research emphasize the superiority of Unet-CBAM as a robust solution for accurate brain tumor classification with BraTS and CE-MRI datasets. By surpassing the performance of Unet, Otsu, and Wishart methods, Unet-CBAM demonstrates its potential to revolutionize the field of medical image analysis. As technology continues to evolve, the integration of deep learning techniques and attention mechanisms, as demonstrated by Unet-CBAM, holds significant promise in addressing complex challenges and advancing patient care within the realm of medical imaging.

Declaration

• Financial Support: The authors confirm that we did not receive any financial support from any organization for the research presented in this work.
• Ethical statement: Throughout the development of this paper, we maintained strict adherence to ethical guidelines and practices, thereby safeguarding the integrity and credibility of our research.
• Data availability statement: Publicly available datasets [41][42].
• Conflict of Interest: The corresponding author, on behalf of all authors, declares the absence of any conflicts of interest.

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